

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

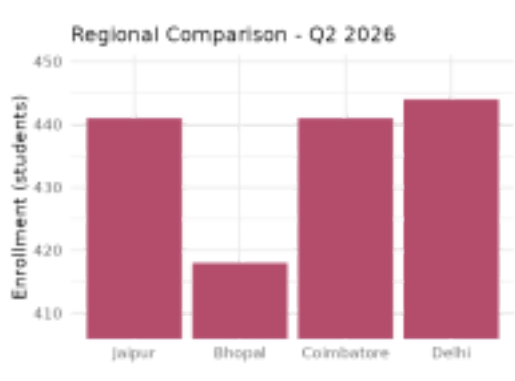
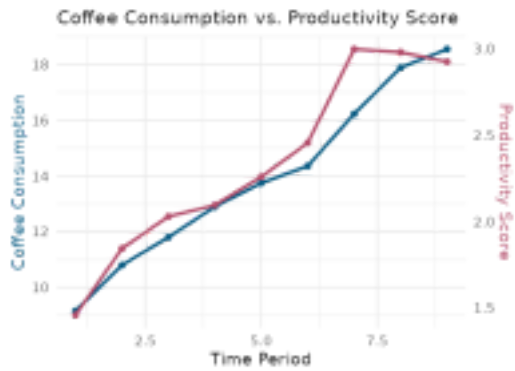


Chart 1

Budget Allocation by Department (12 categories)



Chart 2



3 Chart 4



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

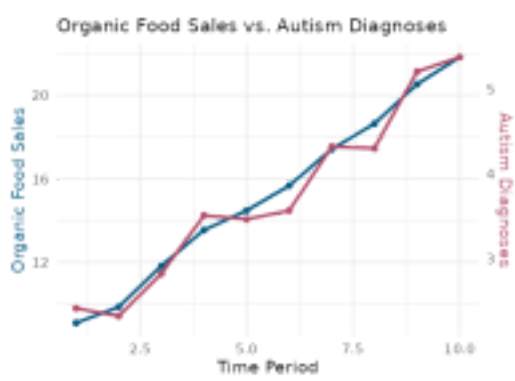


Chart 1

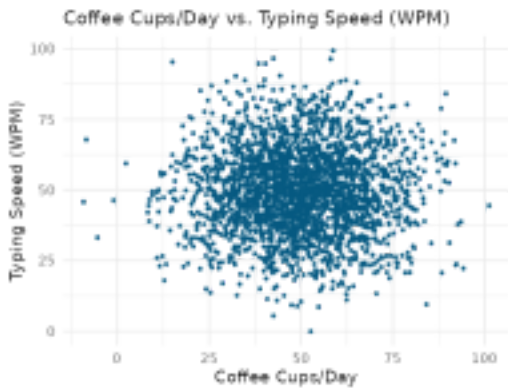
Budget Allocation by Department (11 categories)



Chart 2



3 Chart 4



Chart

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 3 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

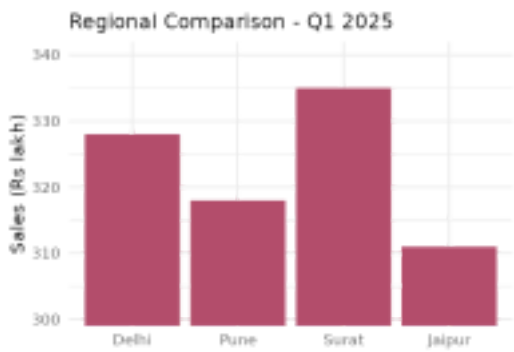
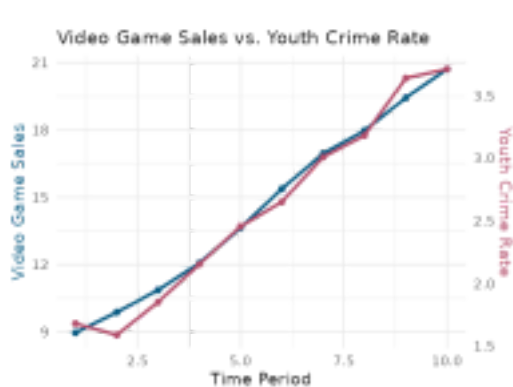


Chart 1

Vote Share by Party (14 categories)



Chart 2



3 Chart 4

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 4 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

Market Share by Product (10 categories)



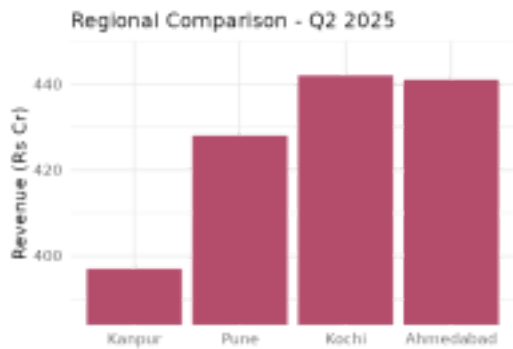
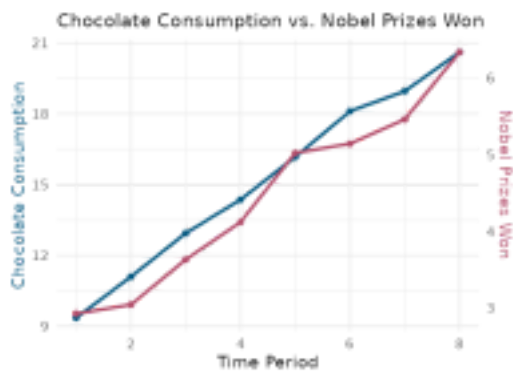
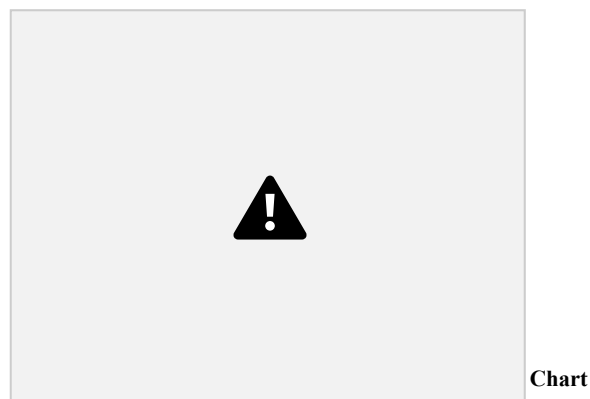


Chart 1

Chart 2



³Chart 4



Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 5 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

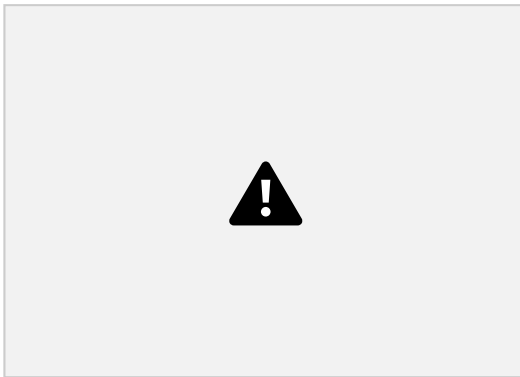


Chart 1

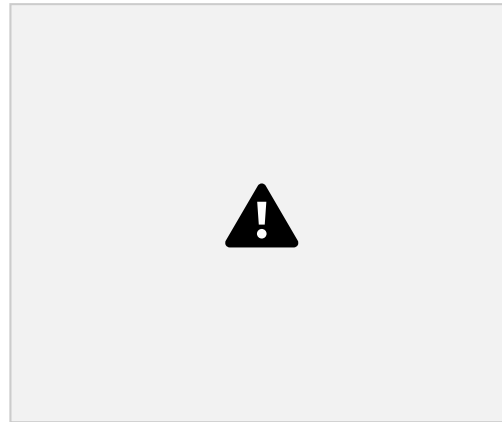
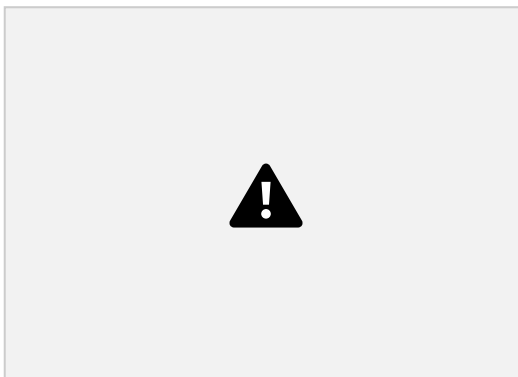
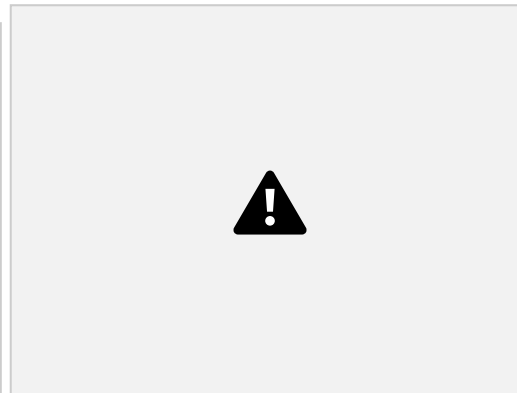


Chart 2



³**Chart 4**



Chart

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Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

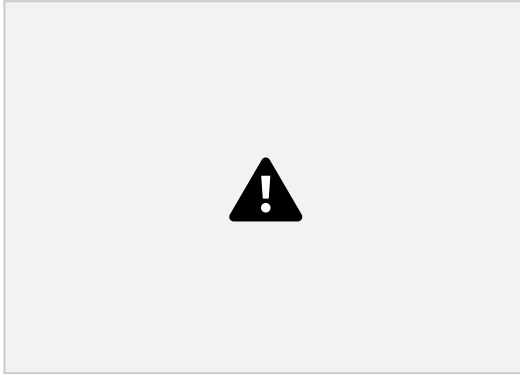


Chart 1

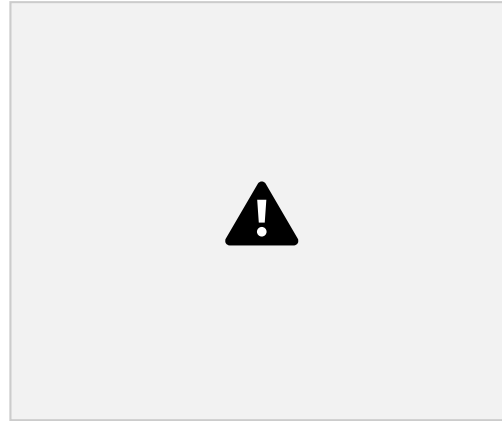
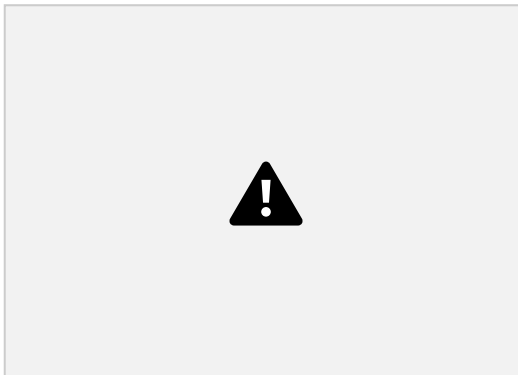
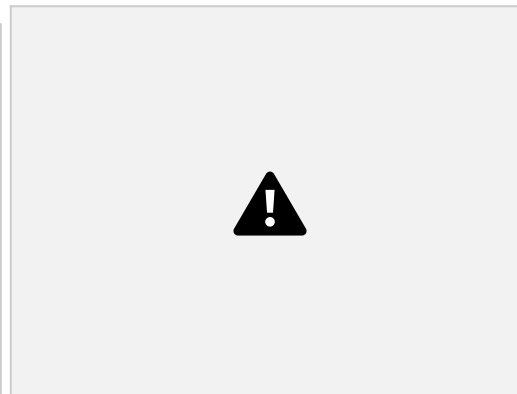


Chart 2



³Chart 4



Chart

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Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

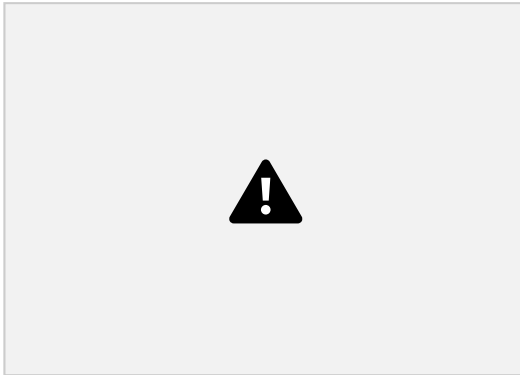


Chart 1

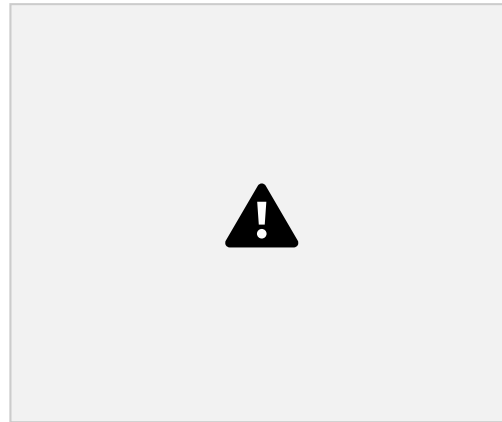


Chart 2

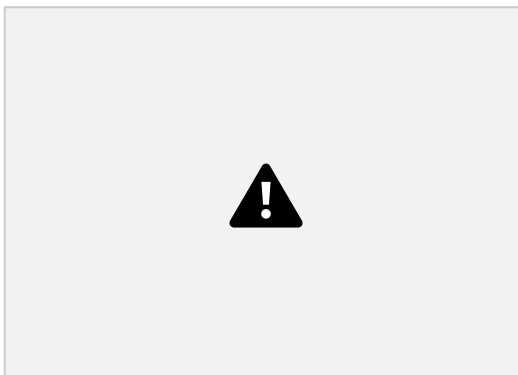


Chart 3

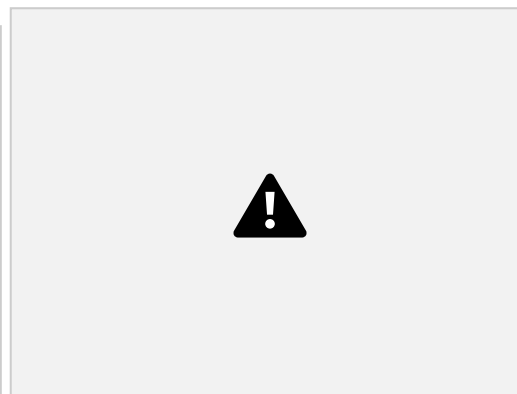


Chart 4

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 8 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

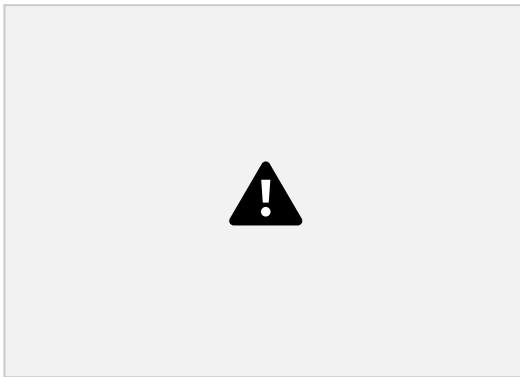


Chart 1

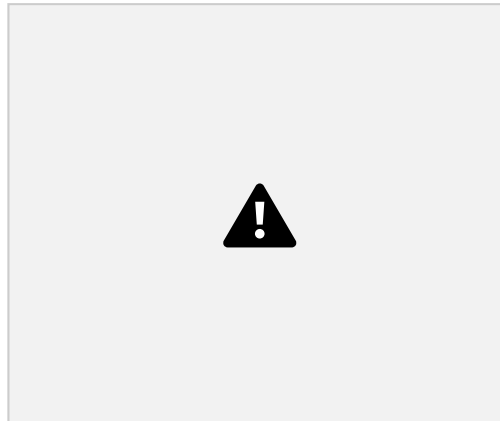
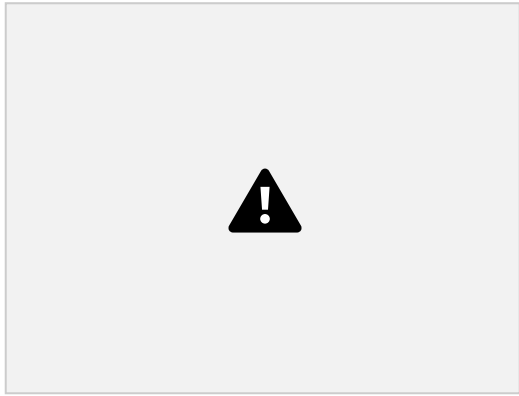
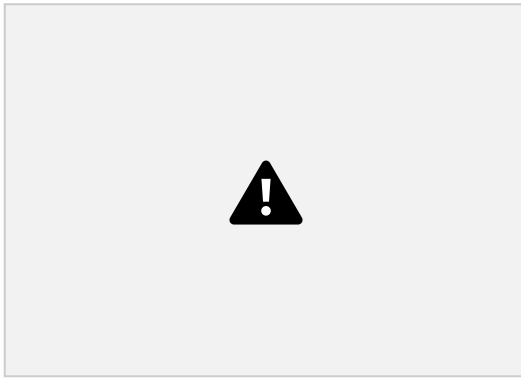


Chart 2



Chart

³Chart 4

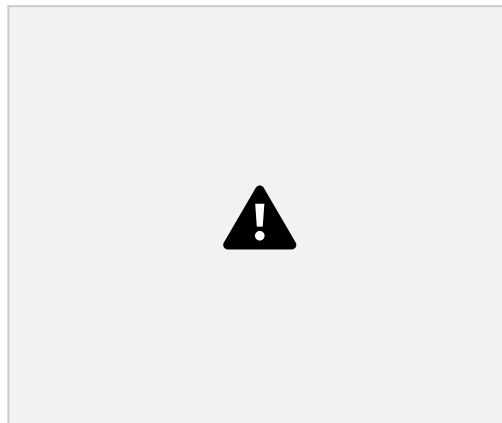
Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 9 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?



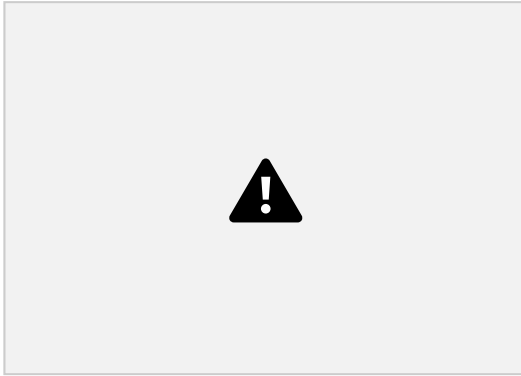
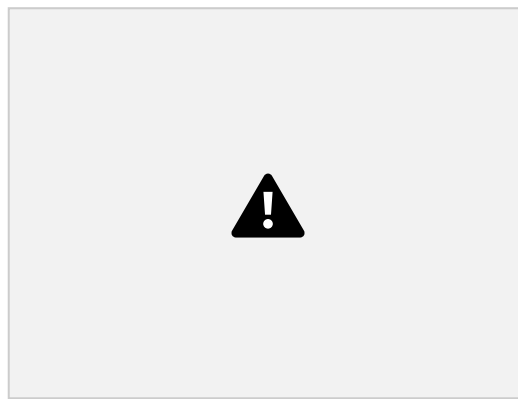
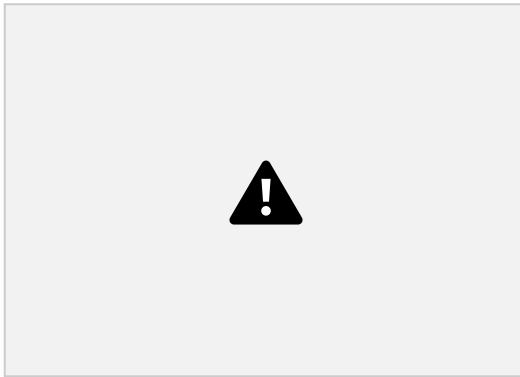


Chart 1

Chart 2



Chart

³Chart 4

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 10 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

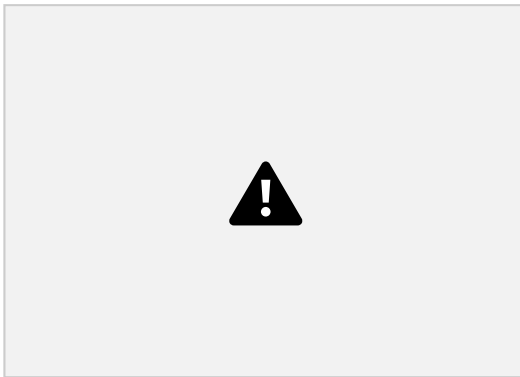


Chart 1

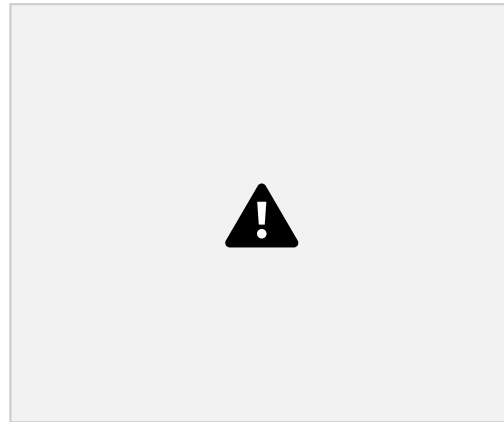
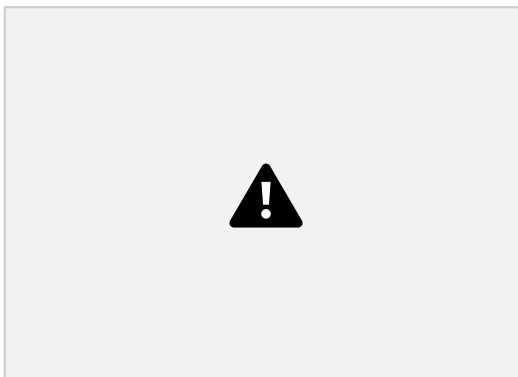
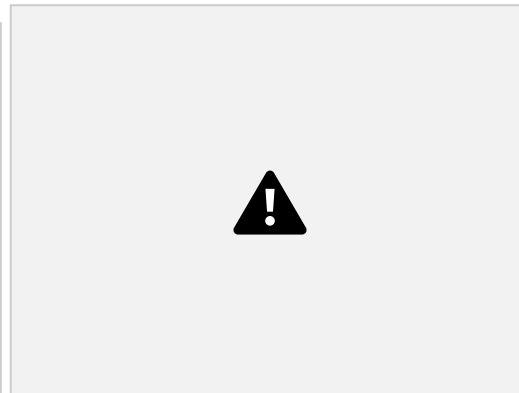


Chart 2



³**Chart 4**



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

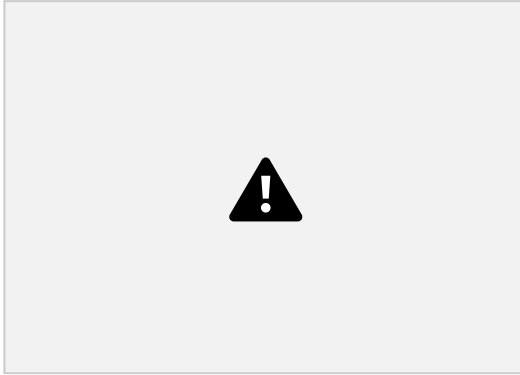


Chart 1

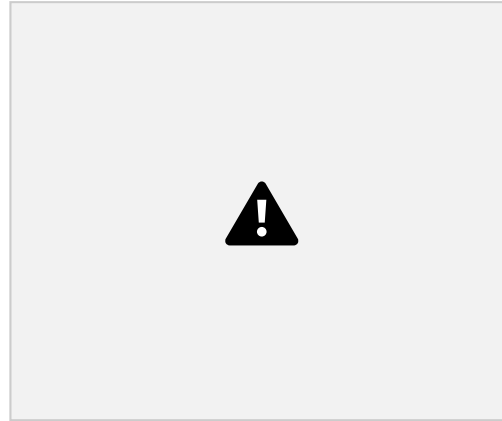
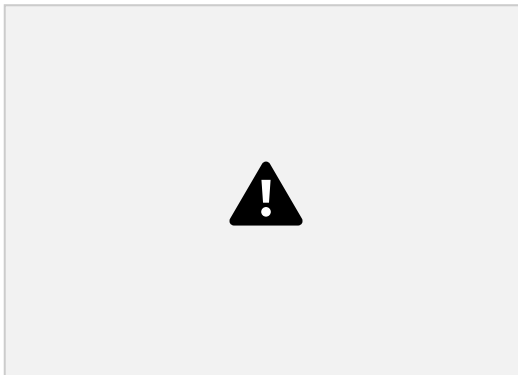
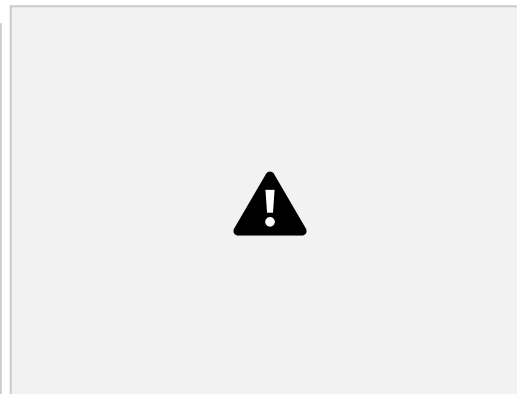


Chart 2



³Chart 4



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

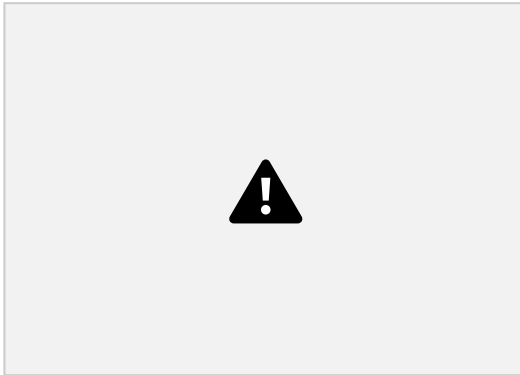


Chart 1

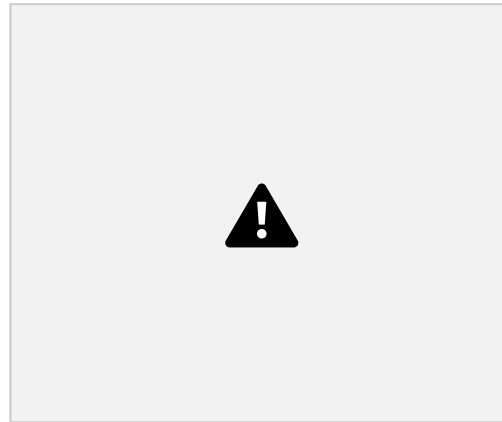


Chart 2

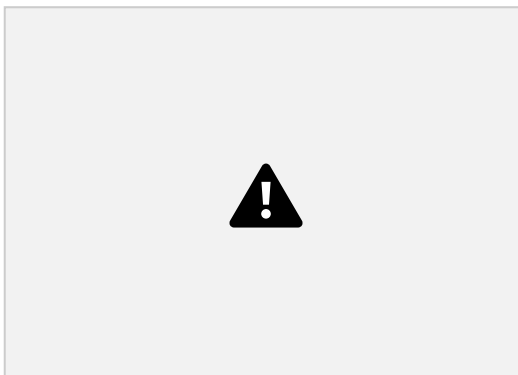


Chart 3

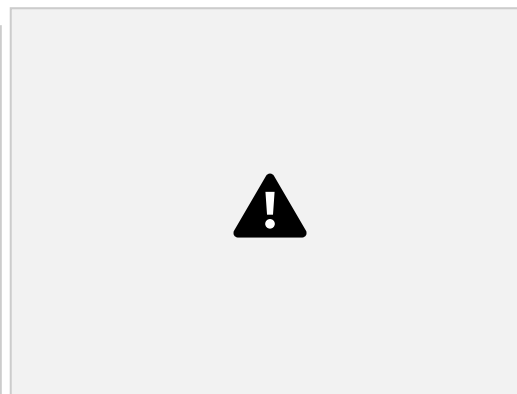


Chart 4

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 13 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

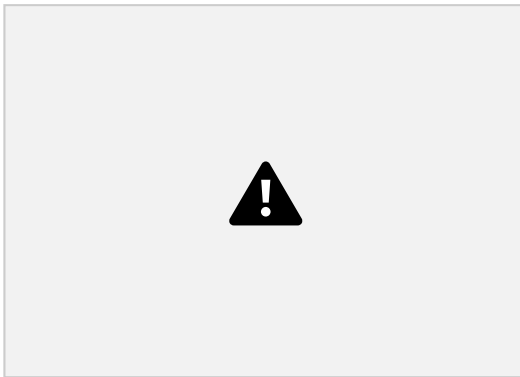


Chart 1

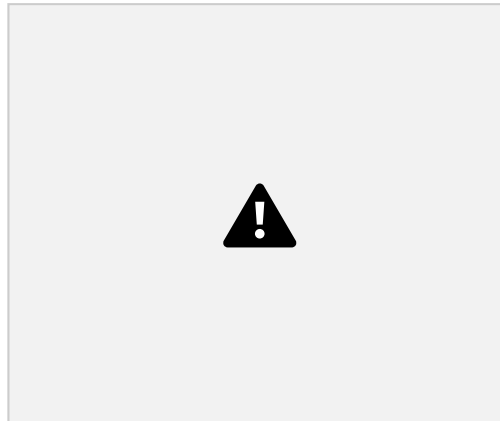
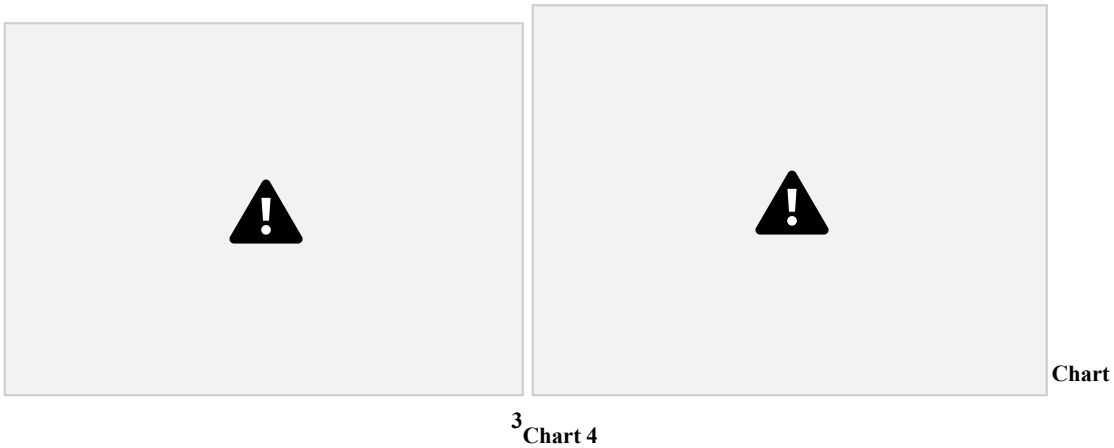


Chart 2



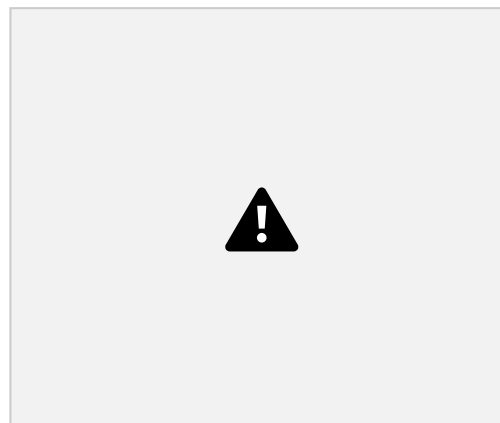
Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 14 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?



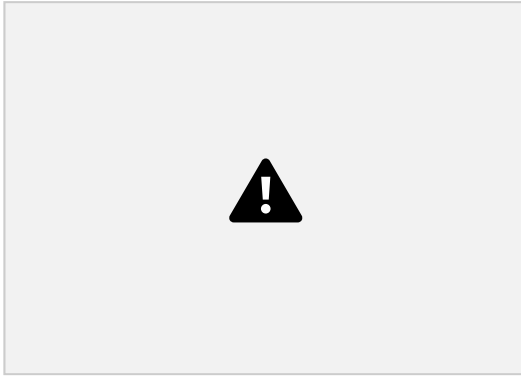
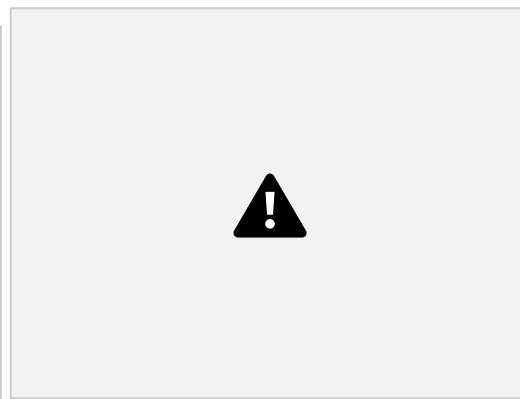
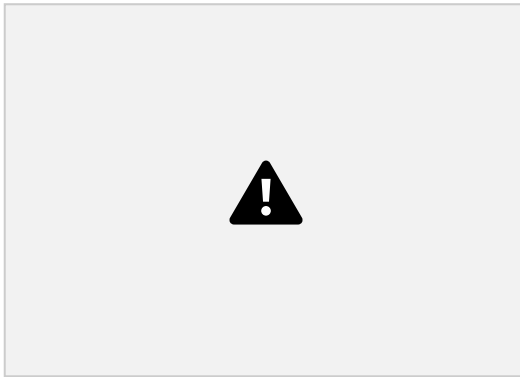


Chart 1

Chart 2



Chart

³Chart 4

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 15 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

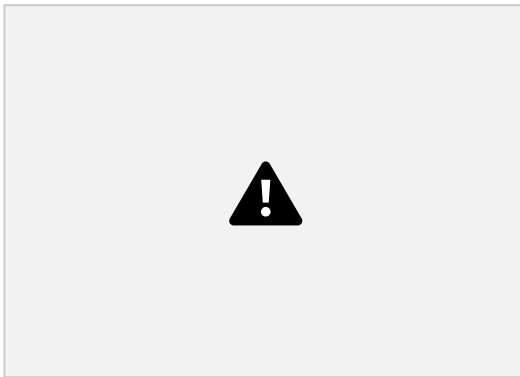


Chart 1

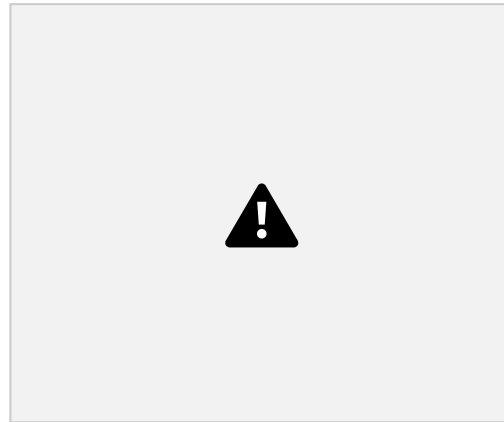
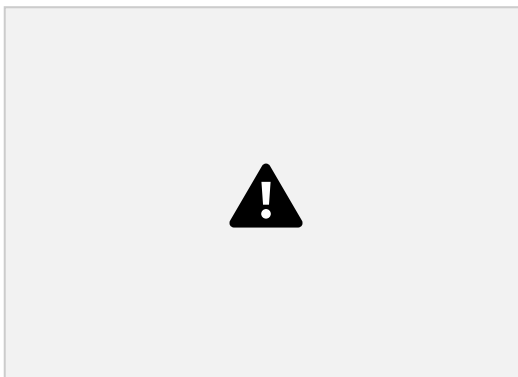
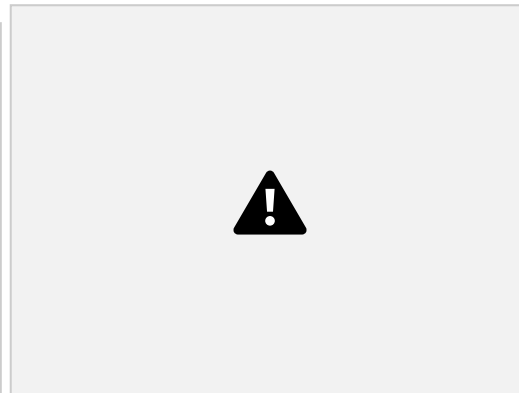


Chart 2



³**Chart 4**



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

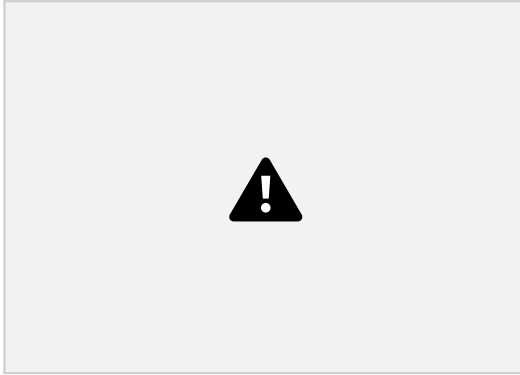
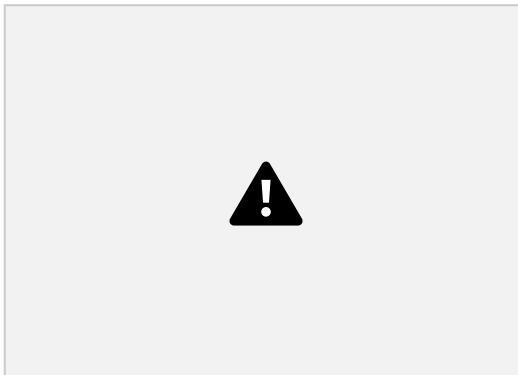


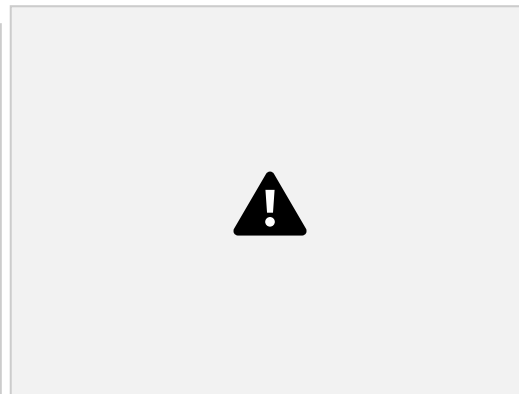
Chart 1



Chart 2



³Chart 4



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

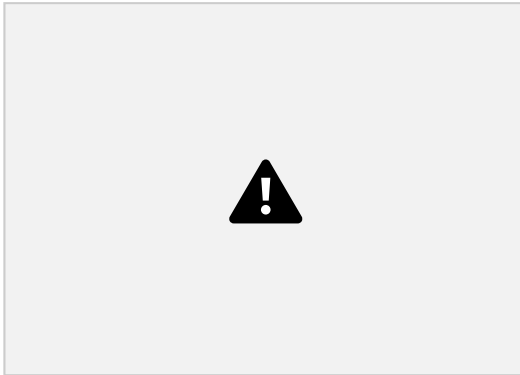


Chart 1

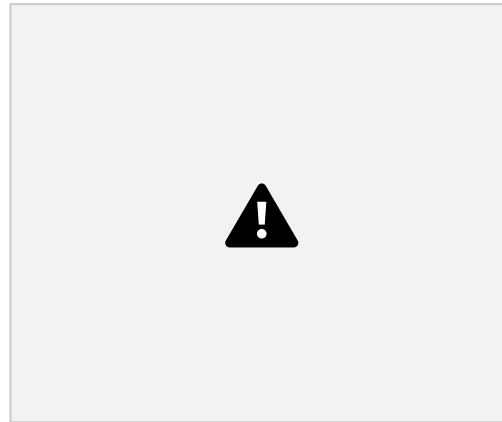
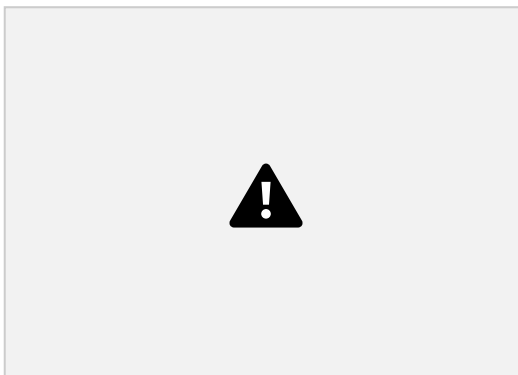
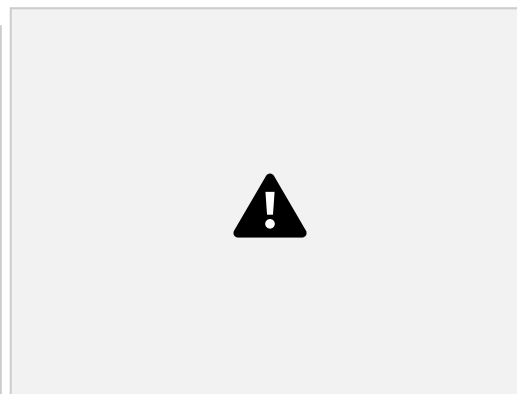


Chart 2



³Chart 4



Chart

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 18 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

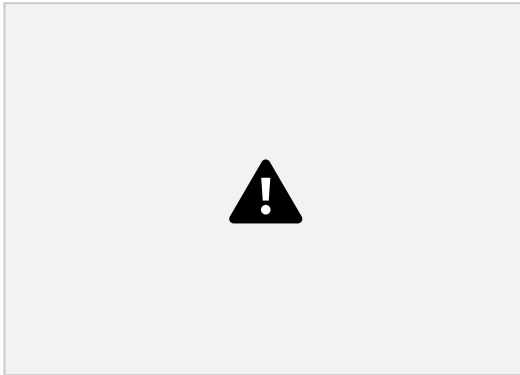


Chart 1

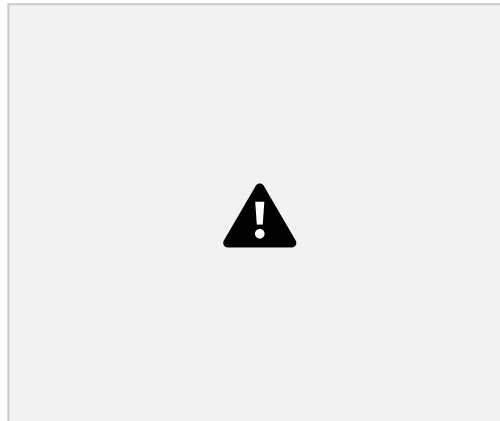


Chart 2

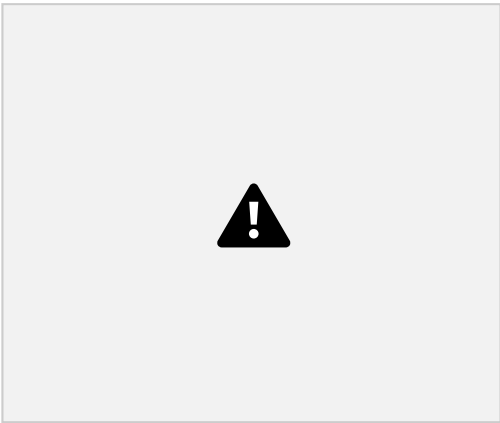


Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?



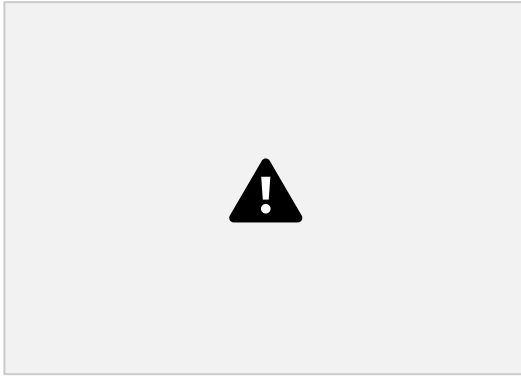
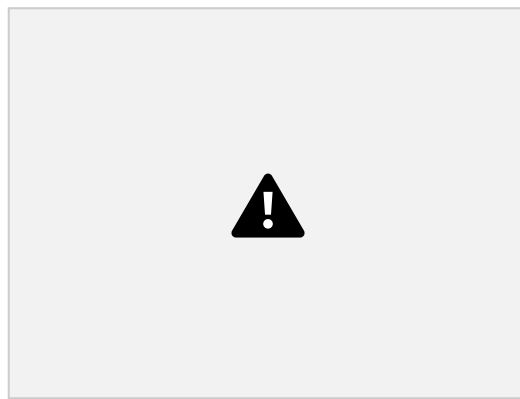
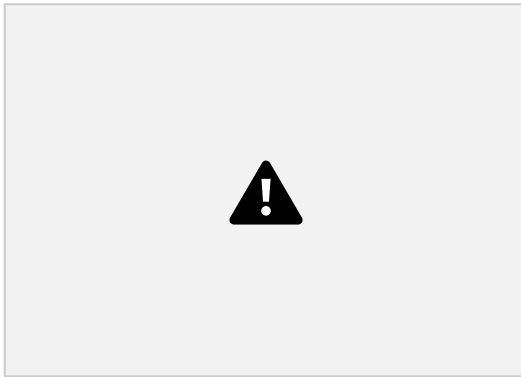


Chart 1

Chart 2



Chart

³Chart 4

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 20 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

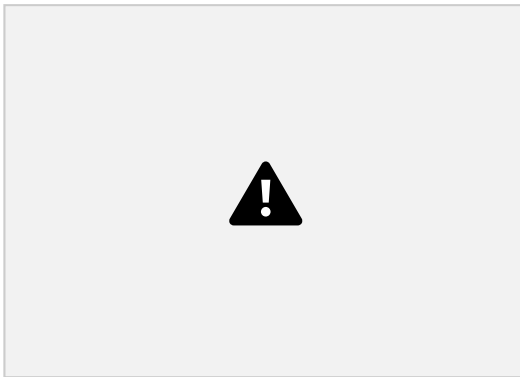


Chart 1

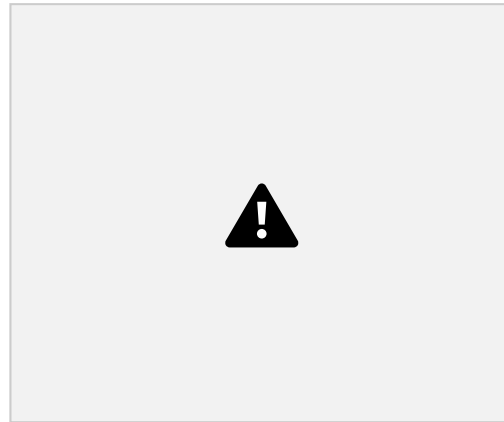
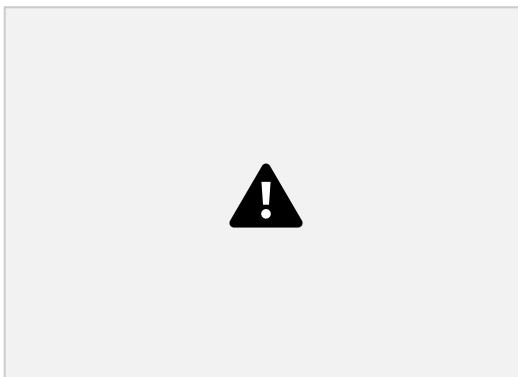
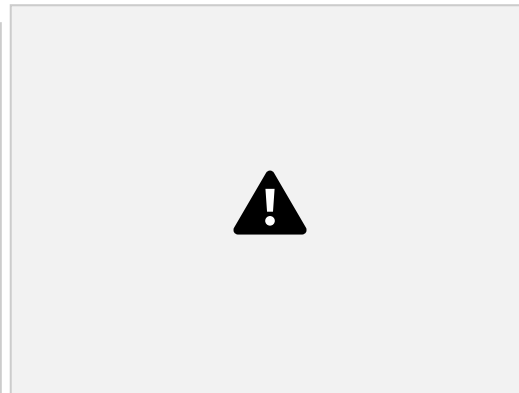


Chart 2



³**Chart 4**



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

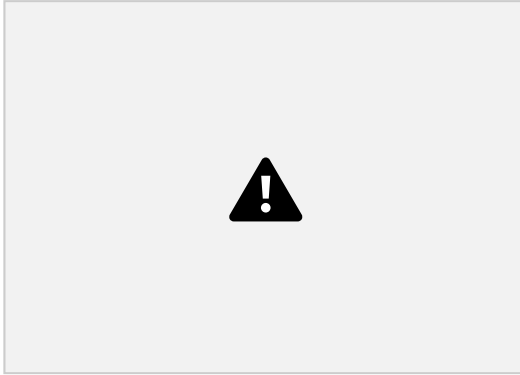
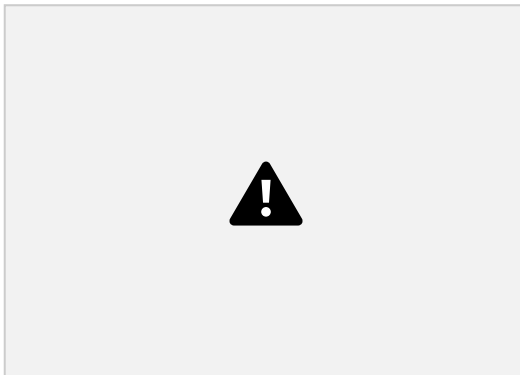


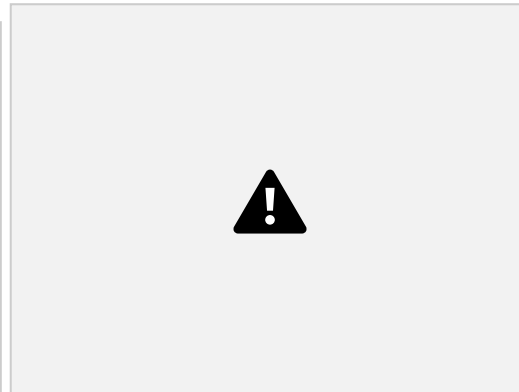
Chart 1



Chart 2



³Chart 4



Chart

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Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

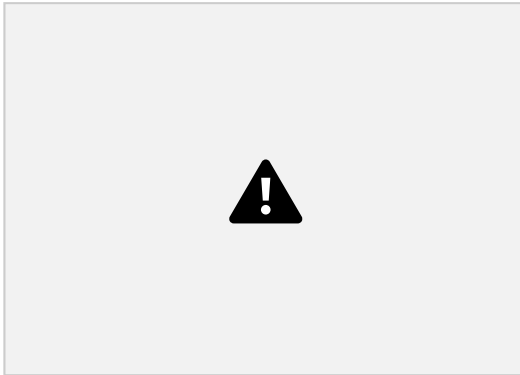


Chart 1

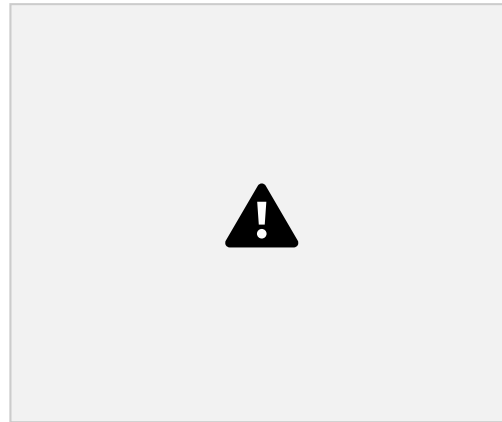
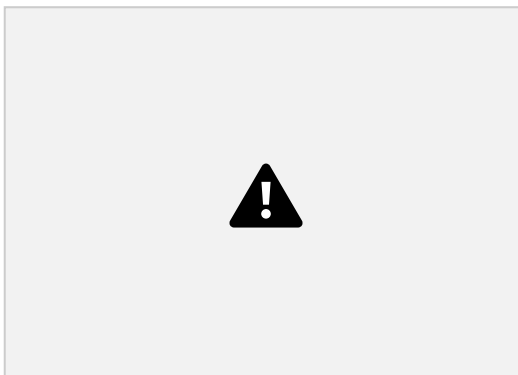
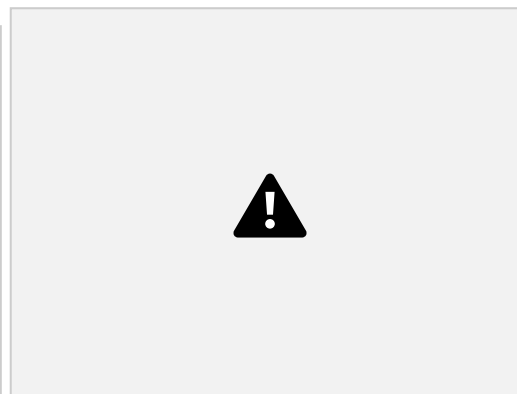


Chart 2



³Chart 4



Chart

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 23 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

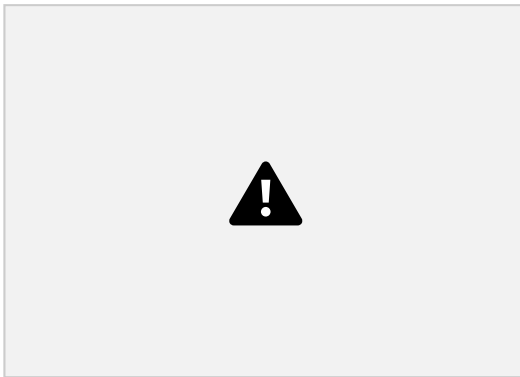


Chart 1

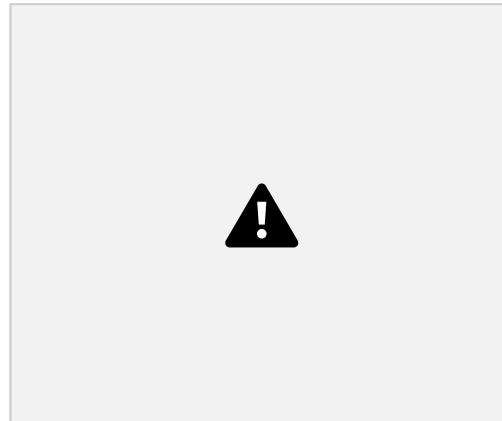


Chart 2



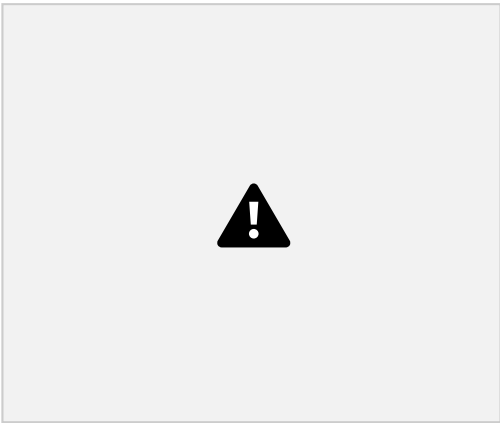
Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 24 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?



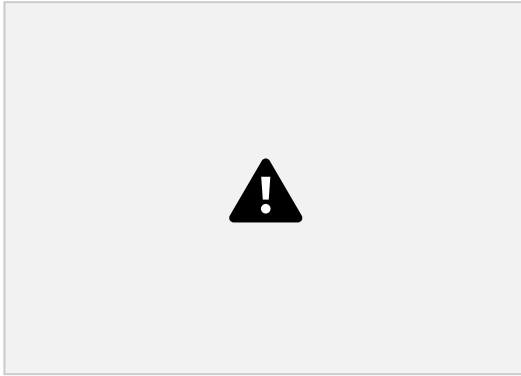
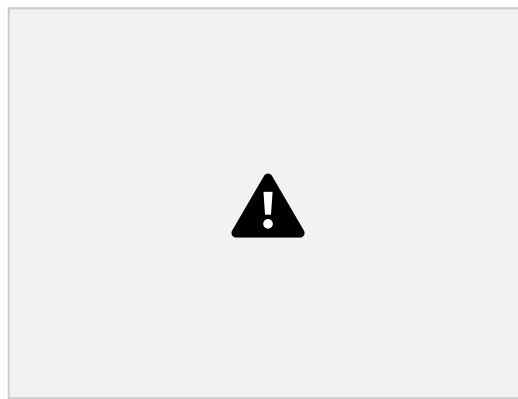
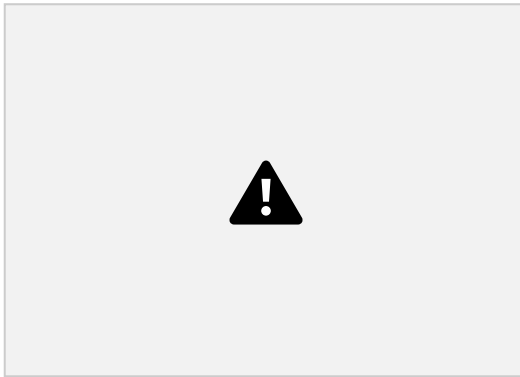


Chart 1

Chart 2



Chart

³Chart 4

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 25 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

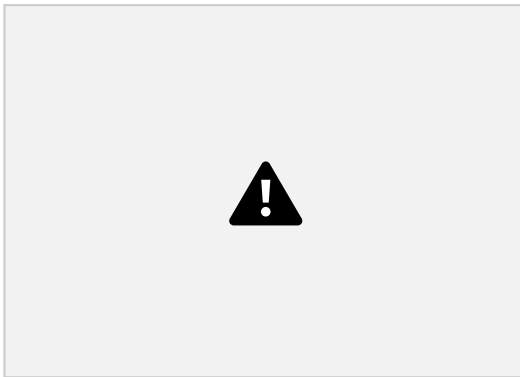


Chart 1

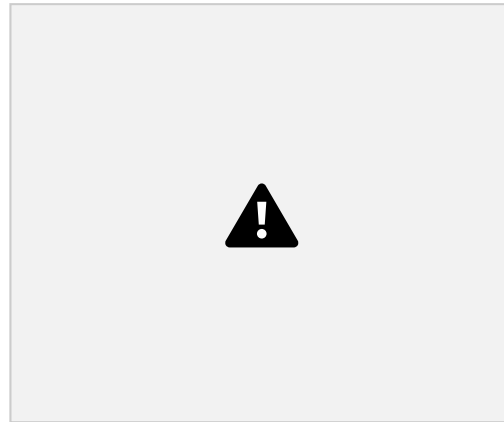
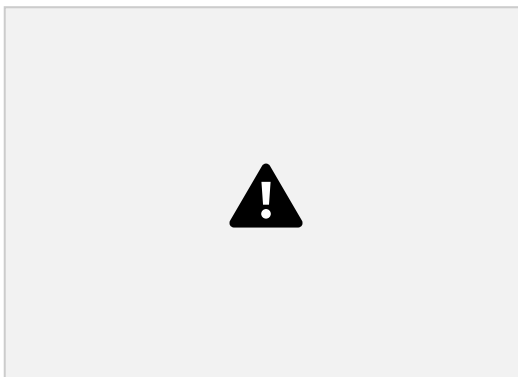
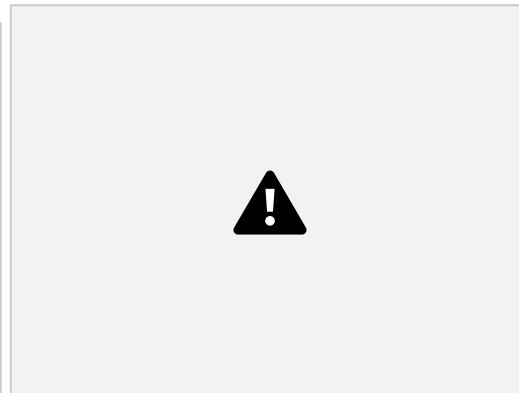


Chart 2



³**Chart 4**



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

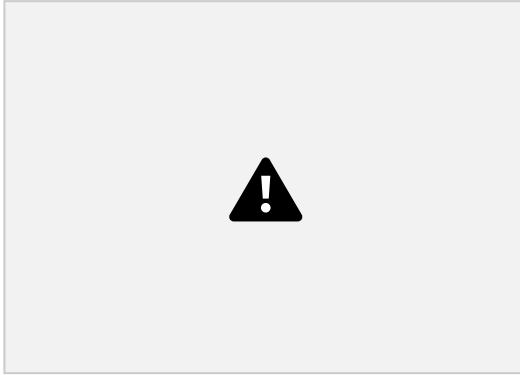


Chart 1

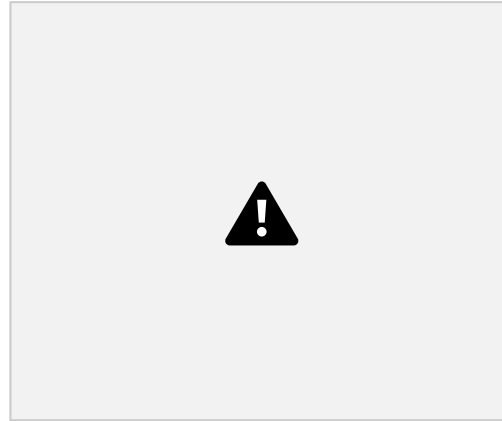
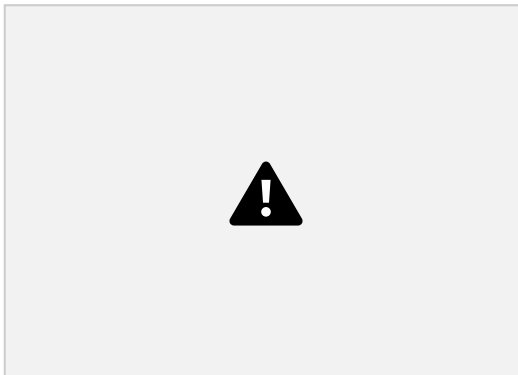
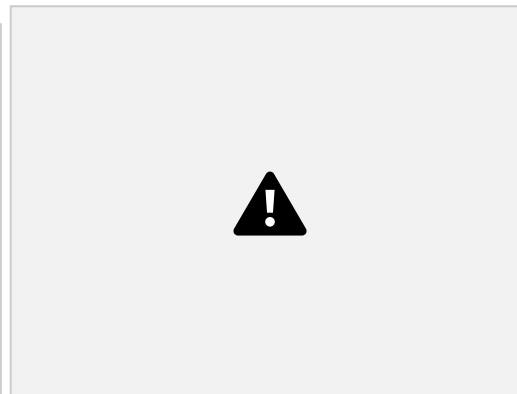


Chart 2



³Chart 4



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

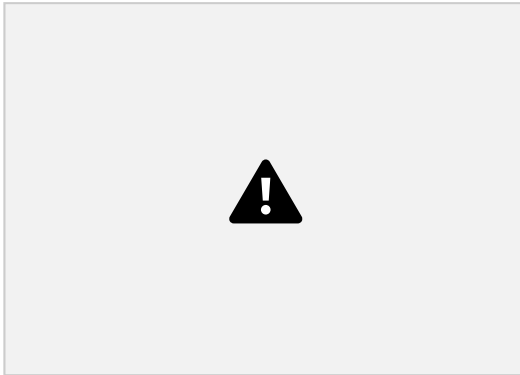


Chart 1

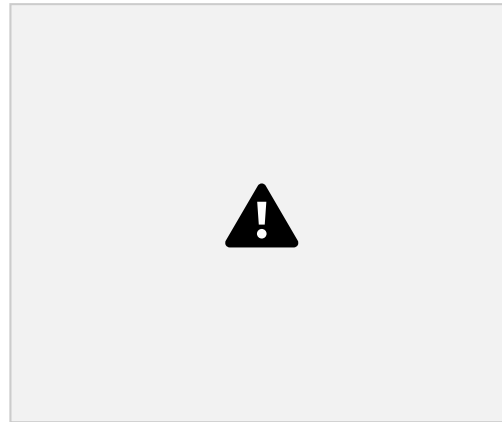
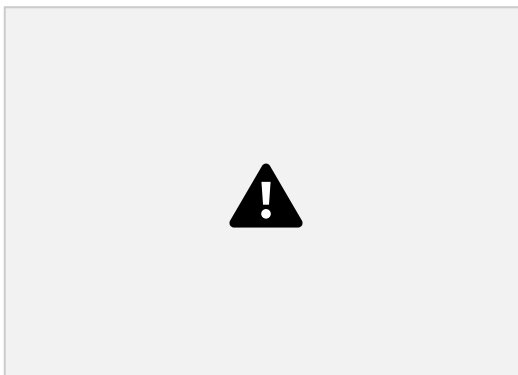
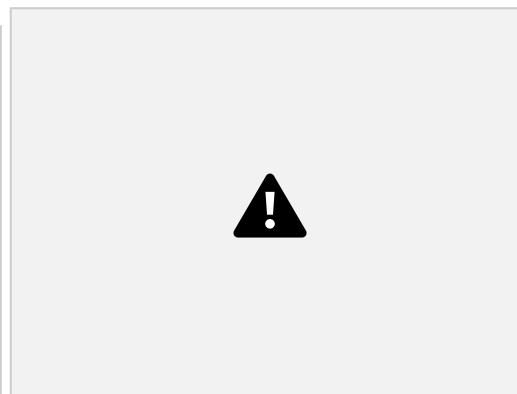


Chart 2



³Chart 4



Chart

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 28 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

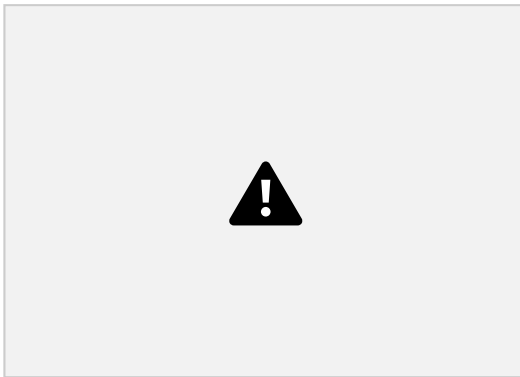


Chart 1

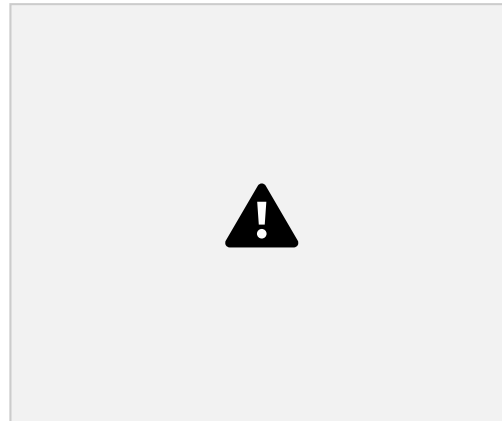
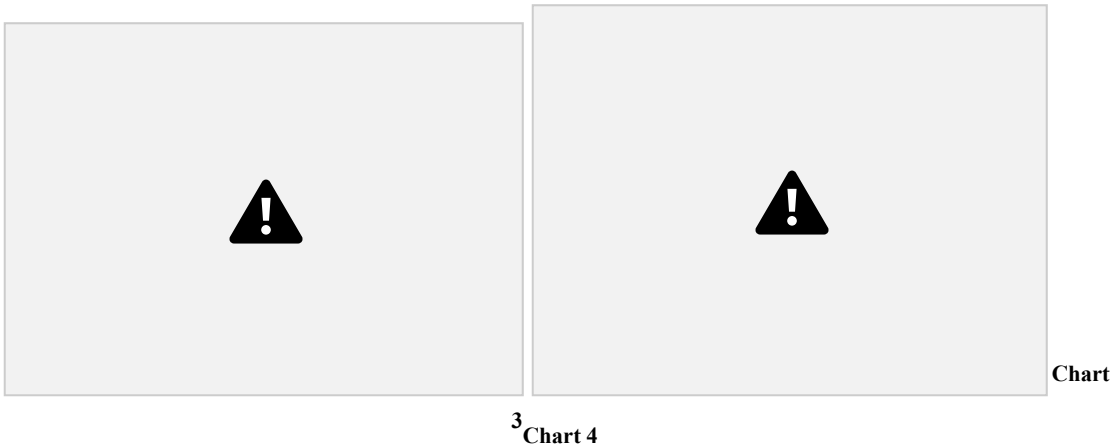


Chart 2



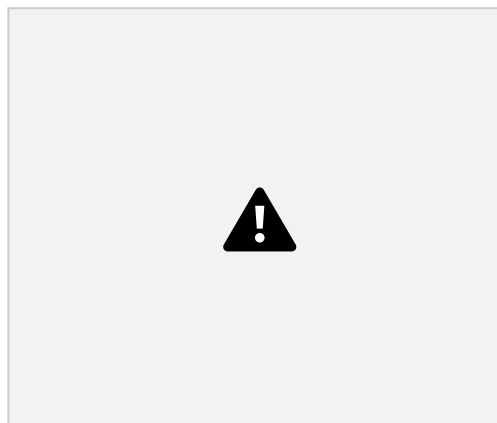
Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 29 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?



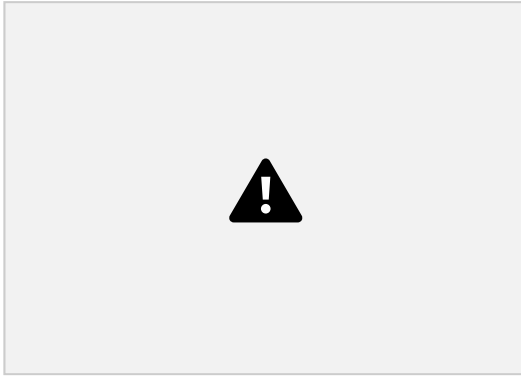
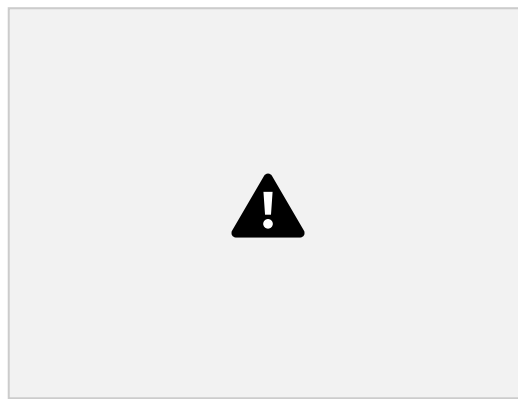
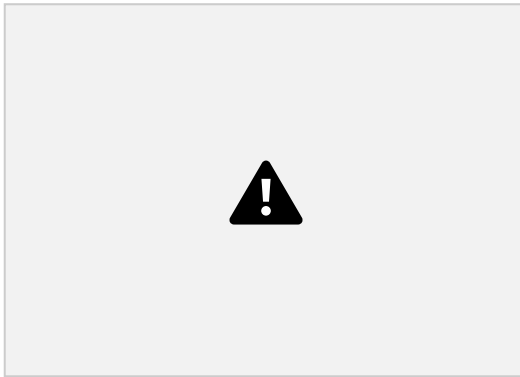


Chart 1

Chart 2



Chart

³Chart 4

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 30 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

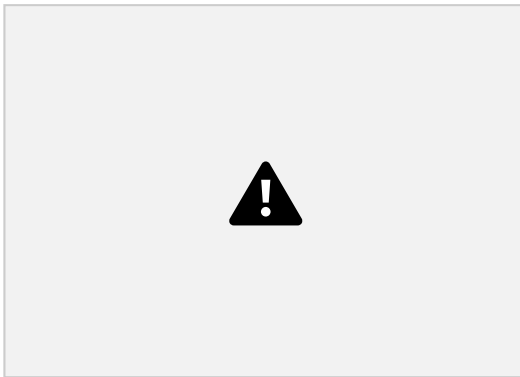


Chart 1

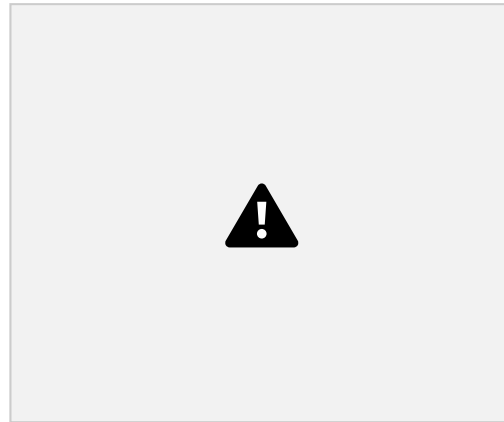
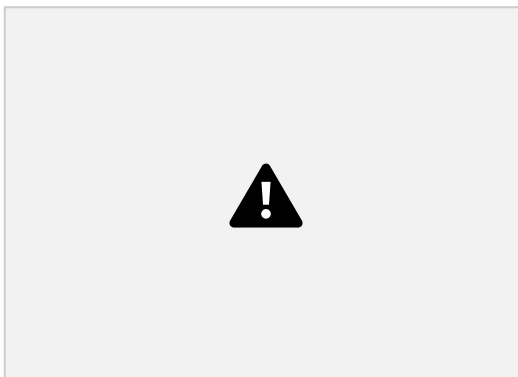
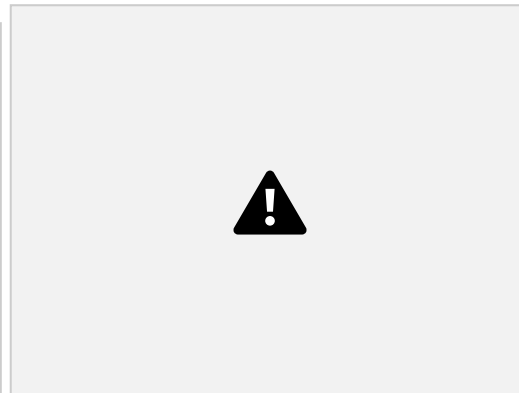


Chart 2



³**Chart 4**



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

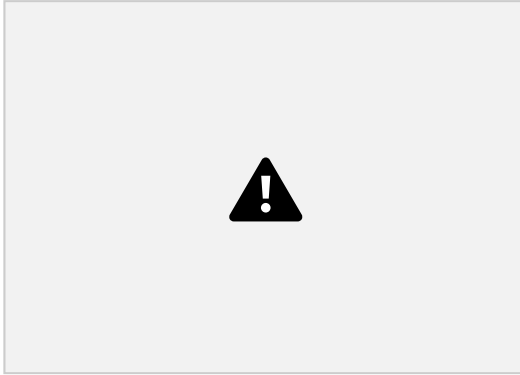
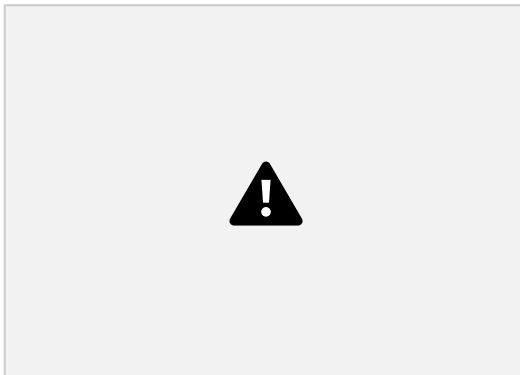


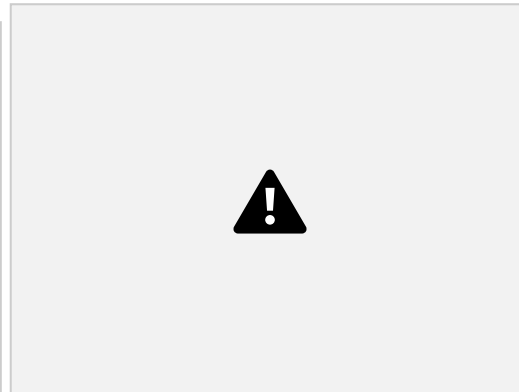
Chart 1



Chart 2



³Chart 4



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

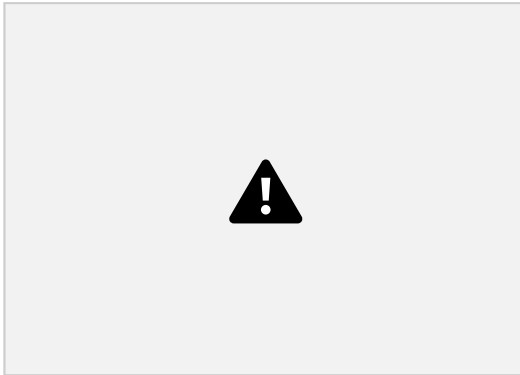


Chart 1

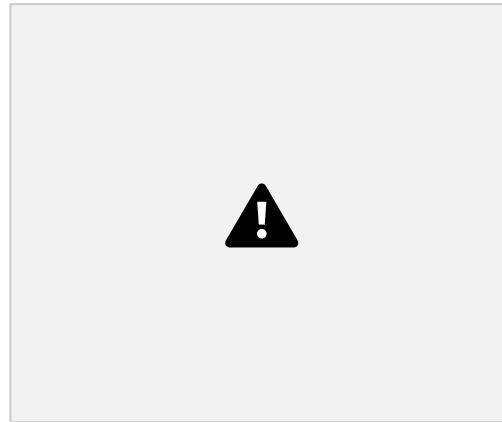
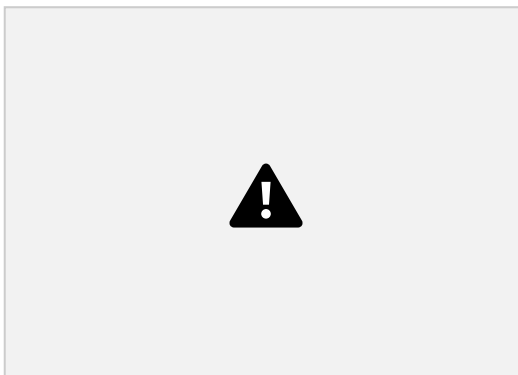
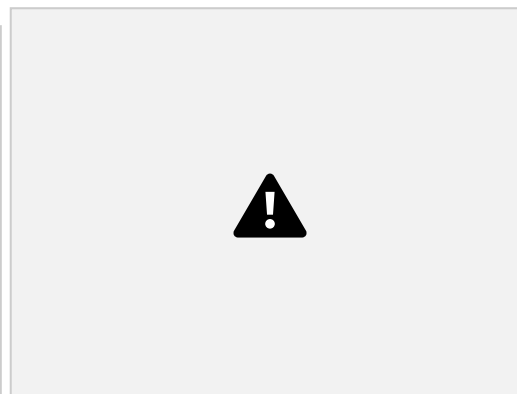


Chart 2



³Chart 4



Chart

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 33 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

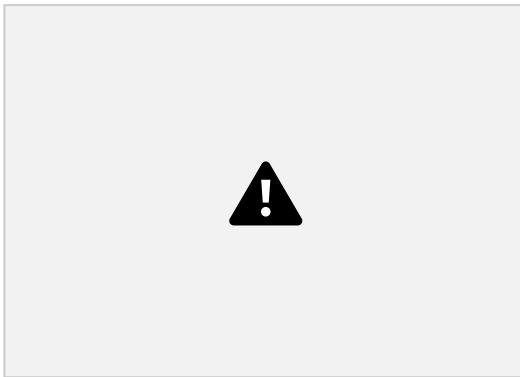


Chart 1

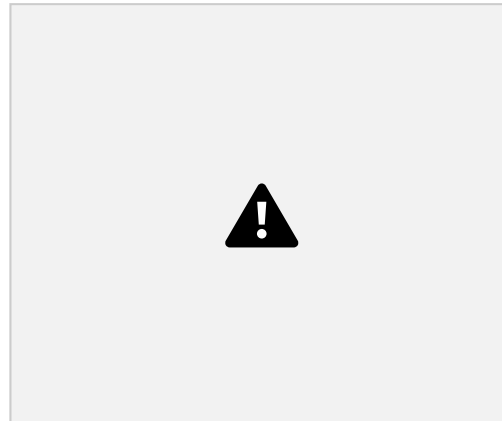
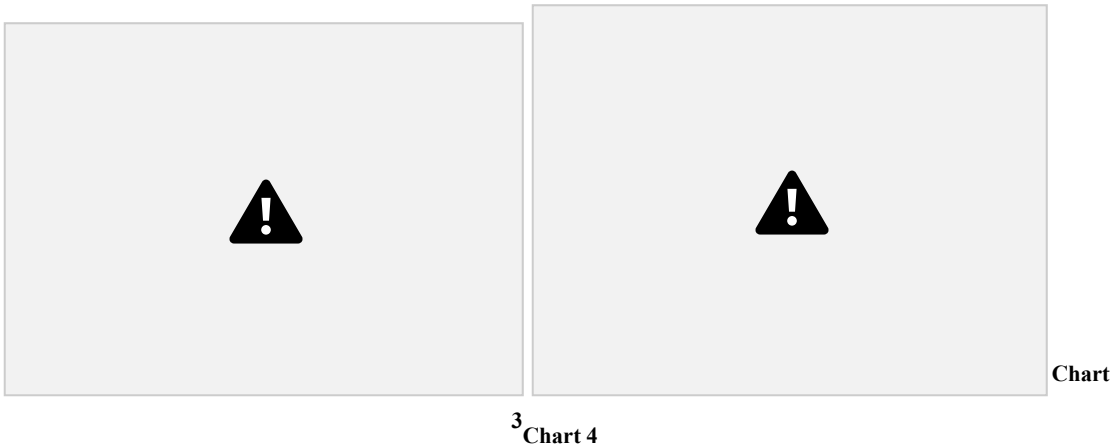


Chart 2



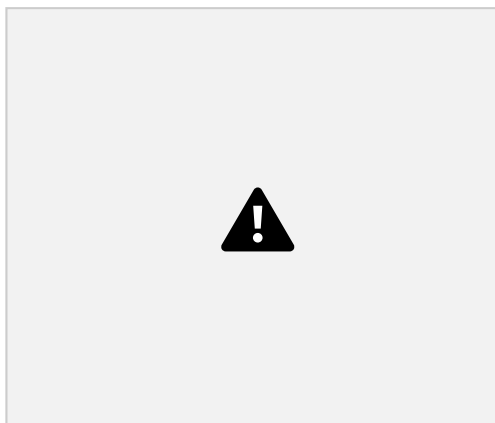
Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 34 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?



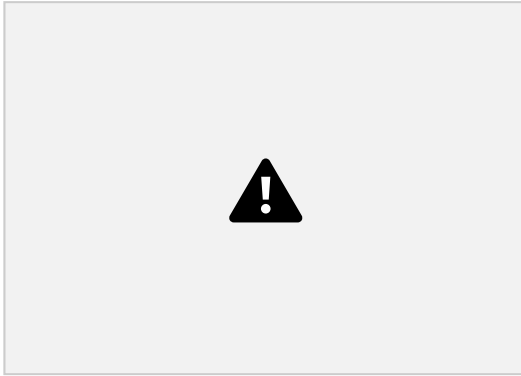
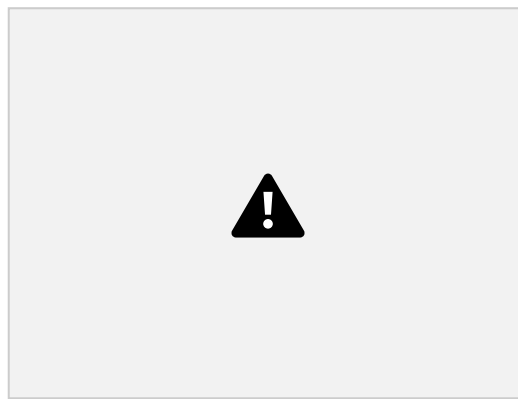
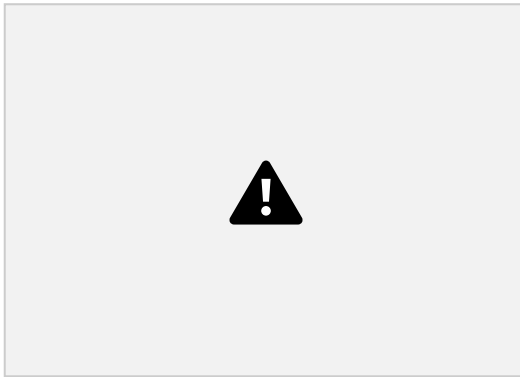


Chart 1

Chart 2



Chart

³Chart 4

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 35 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

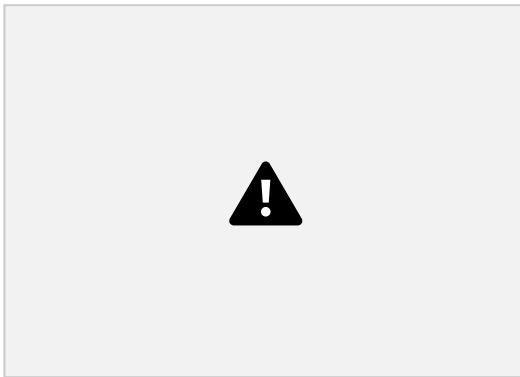


Chart 1

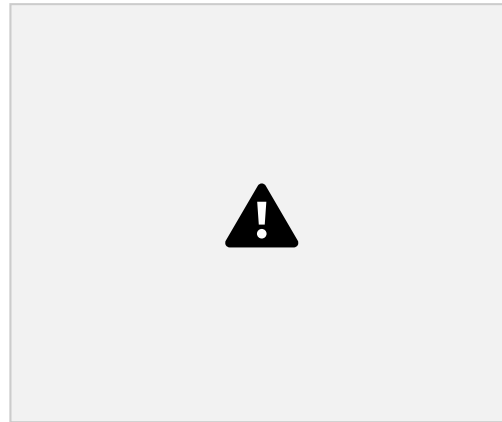
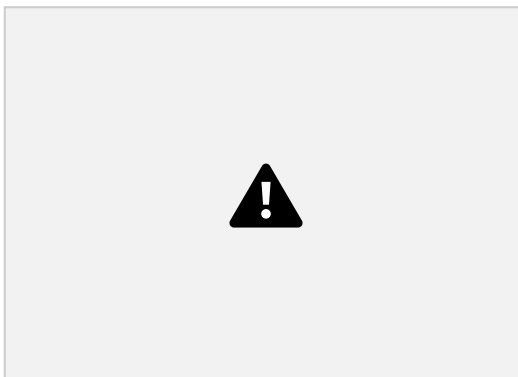
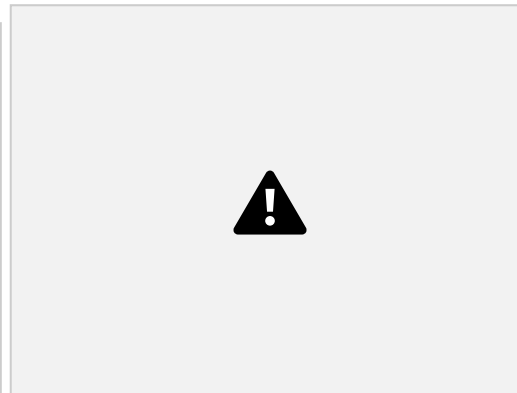


Chart 2



³**Chart 4**



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

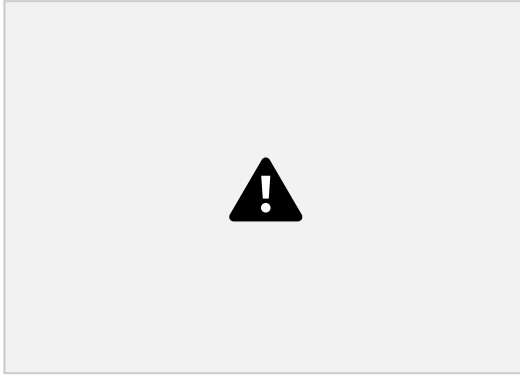


Chart 1

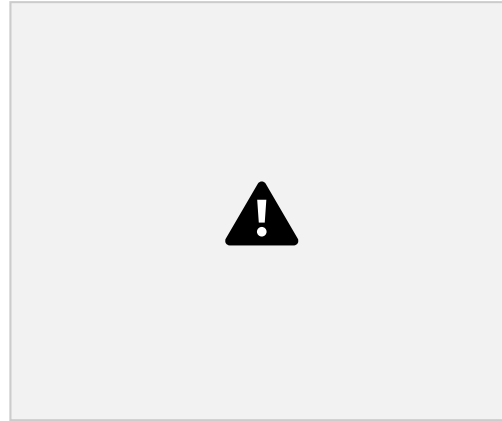
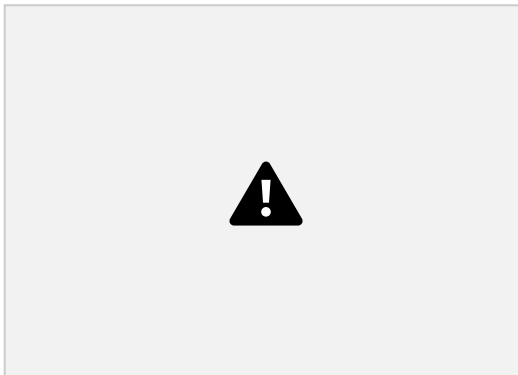
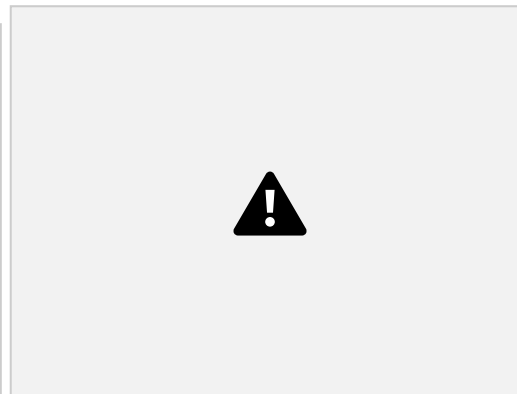


Chart 2



³Chart 4



Chart

Your notes:

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

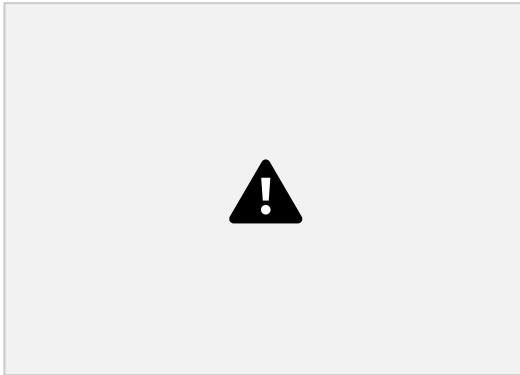


Chart 1

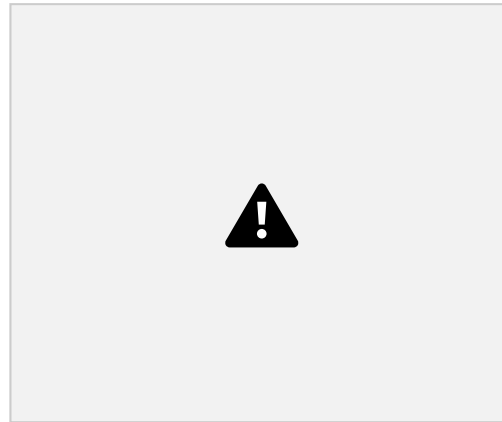
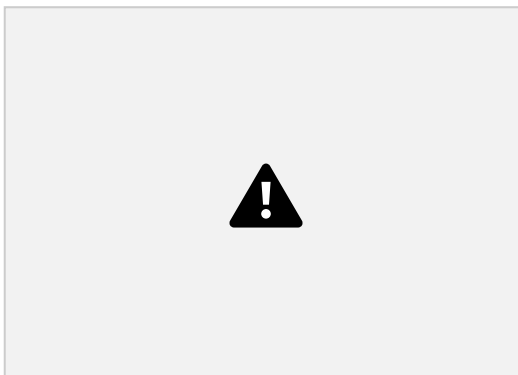
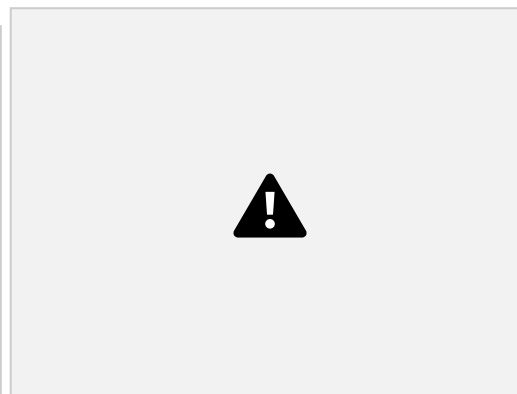


Chart 2



³Chart 4



Chart

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 38 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

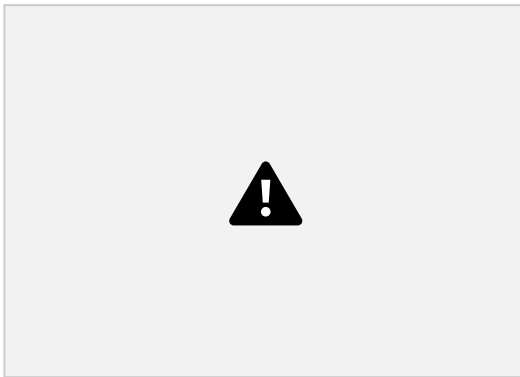


Chart 1

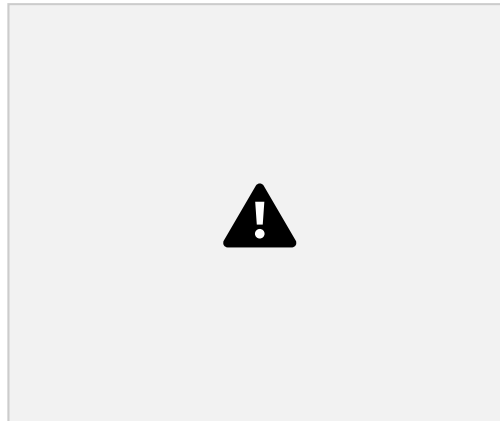
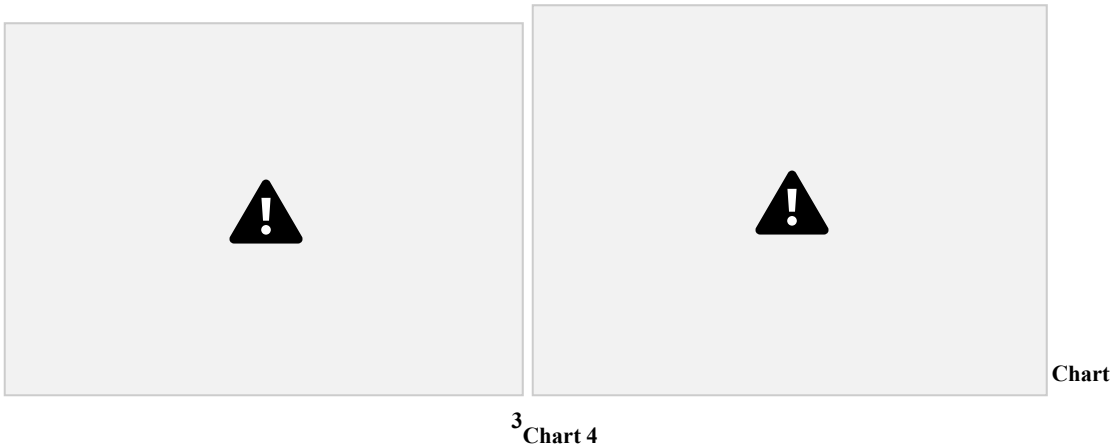


Chart 2



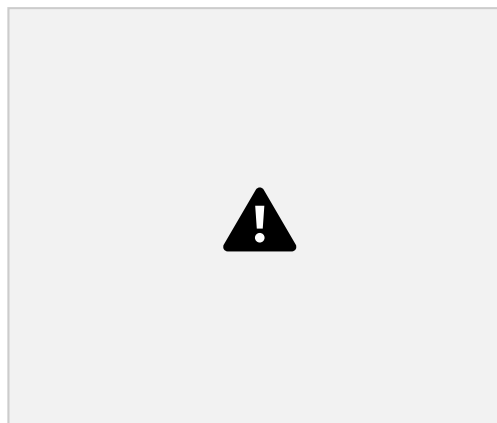
Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 39 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?



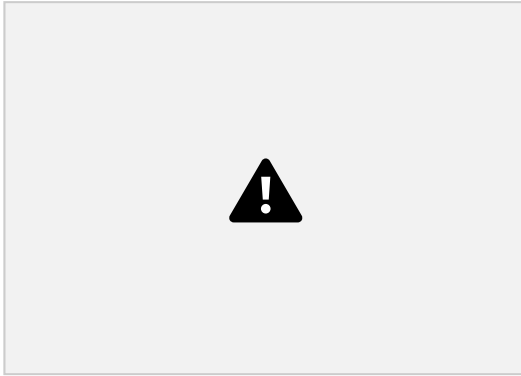
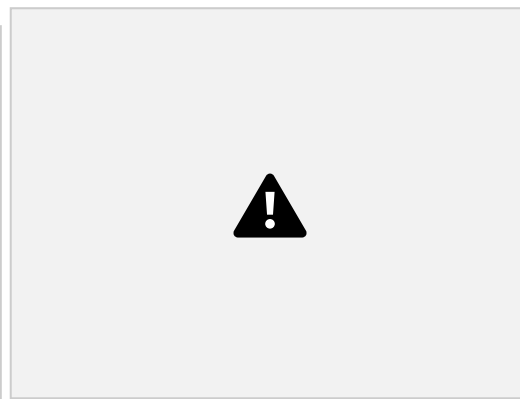
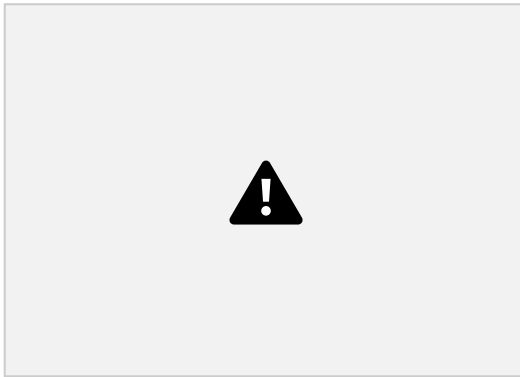


Chart 1

Chart 2



Chart

³Chart 4

Your notes:

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (Set 40 of 40 — every set uses different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

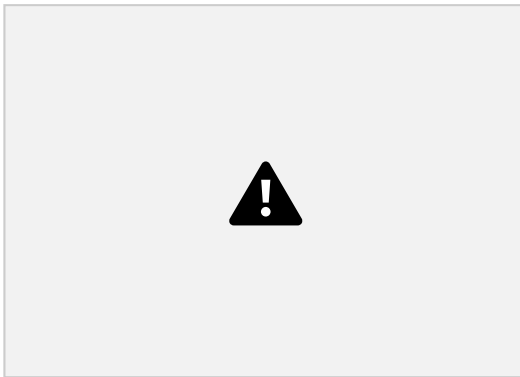


Chart 1

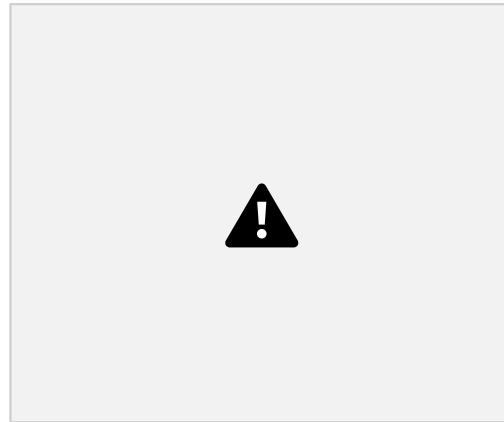
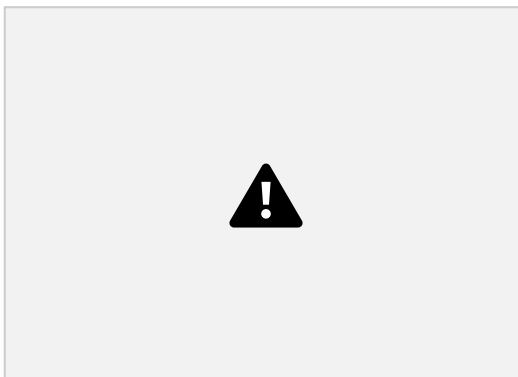
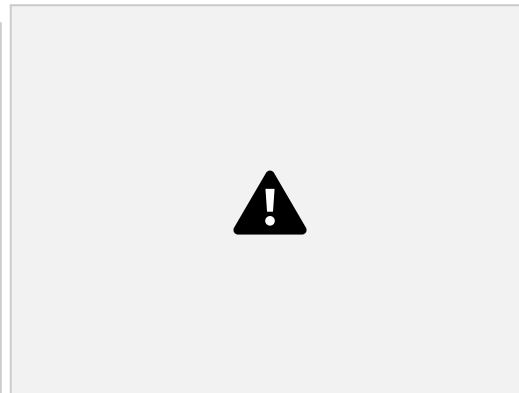


Chart 2



³**Chart 4**



Chart

Your notes:

Answer Key (Instructor Only — do not photocopy this page)

Every one of the 40 sets uses different cities, categories, metrics and random data — so no two students can compare answers directly. But every set repeats the same 4 underlying chart problems, explained generically below. The specific numbers will differ on each student's sheet; the lesson is the same.

Chart 1 — Truncated y-axis (bar chart)

Every student's Chart 1 is a bar chart whose y-axis does NOT start at zero. This inflates small real differences into what looks like a huge gap between bars. Ask the student to read off the y-axis starting value — if it isn't 0, that's the issue. Rule: bars on a linear scale must always start at zero (Principle of Proportional Ink).

Chart 2 — Too many categories in a pie chart

Every student's Chart 2 is a pie chart with 9–14 wedges. With that many slices, colors and labels become impossible to compare accurately — nobody can reliably tell which wedge is 3rd largest. Rule: beyond about 6–8 categories, switch to a sorted bar chart, or use direct labels instead of relying on angle/color.

Chart 3 — Dual y-axis, false correlation

Every student's Chart 3 shows two completely different, often absurd, quantities (e.g. storks sighted vs. human births, ice cream sales vs. drowning incidents) plotted on two independent y-axes, scaled so the lines appear to move together. This is a classic illustration of spurious correlation: the visual overlap is created entirely by the choice of axis scaling, not by any real relationship. Rule: be very cautious with dual-axis charts; prefer a single shared scale, two separate panels, or a real correlation statistic.

Chart 4 — Overplotting (scatter plot)

Every student's Chart 4 is a scatter plot with 1,500–2,800 fully opaque points. Dense regions merge into solid blobs and we lose all sense of where the data is truly concentrated. Rule: for large datasets, use partial transparency (alpha), 2D histograms/hexbins, or contour lines.